1. What does the following program print if the language uses
    a. Shallow binding
    b. Deep binding
    c. Ad-hoc binding

```
procedure main(){
    x:integer := 3;
    y:integer := 1;
    procedure multiply(){
        x := x * y;
    }
    procedure second(P: procedure){
        x:integer := 4;
        P();
    }
    procedure first{
        y:integer := 3;
        second(multiply);
    }
    first();
    print(x);
}
```

2. For each of the parameter passing methods what will be the output?
    a. Pass-by-value 5 6 1
    b. Pass-by-reference 5 9 1
    c. Pass-by-value-result 5 13 1
    d. Pass-by-name 8 6 1

```
procedure main(){
    i:integer;
    a:array[1, 2] of integer;

    procedure f(x:integer){
        a[2] := 6;
        i := 1;
        x := x + 3;
    }

    a[1] := 5;
    a[2] := 10;
    i := 2;
    f(a[i]);
    print(a[i], a[2], i)
}
```

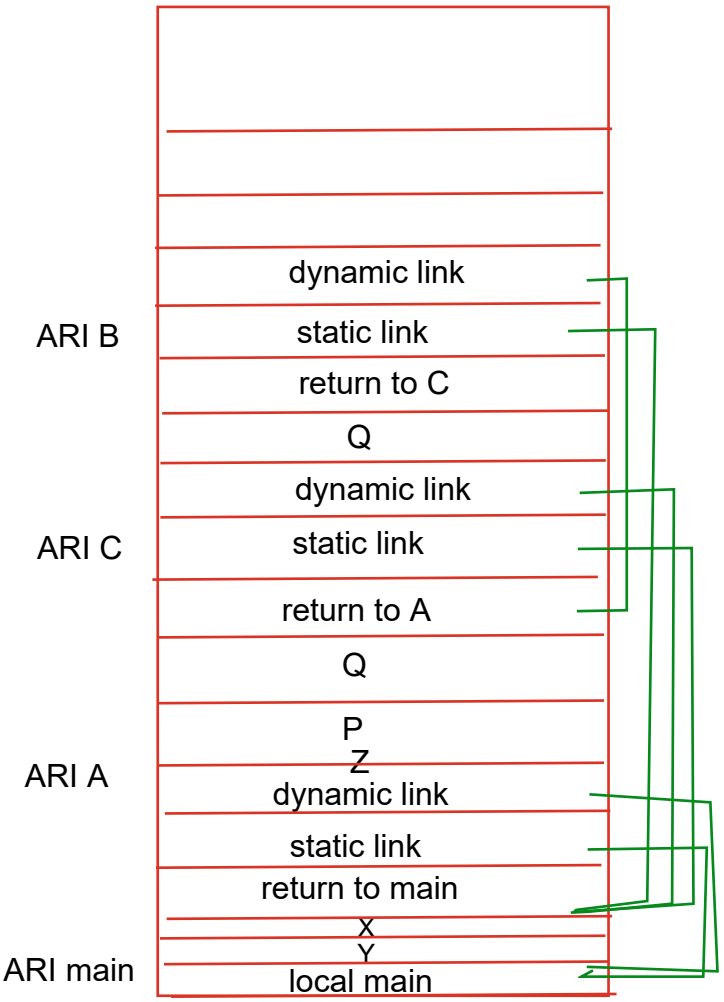
3. What will be the values of s1 and s2 if the operands in the expression are evaluated
    a. Left to right 10 11
    b. Right to left 8 21

```
int sub(int *i){
    *i +=2;
    return (*i) + 3;
}

void main(){
    int i=1, j=1, s1, s2;
    s1 = sub(&i) + (i + 1);
    s2 = (j * 5) + sub(&j);
}
```

4. For the code below, show the stack with all activation record instances when execution reaches position 1 in the following skeletal program.

```
program main {
    int X, Y;
    procedure A(int P, int Q){
        int Z;
        procedure B(){
            ... <--- Position 1
        }
        procedure C(int Q){
            ...
            B();
            ...
        }
        ...
        C(Z);
        ...
    }
    ...
    A(X, Y);
    ...
}
```



5. Fill in the empty parts (there are 5 parts) of the following Prolog program to get the given output for the query.

   a. **Query**

      solve(*A*,B,*C*).

   b. **Output**

      A=3, B=4, *C*=5
      A=c, B=k, *C*=d
      A=c, B=k, *C*=m
      A=c, B=i, *C*=d
      A=c, B=i, *C*=m

```
S(d).
S(m).
t(k).
t(i).
d(c).
d(j).
solve(_____).
solve(X,Y,Z):-___(X), !, ___(Y), !, ___(Z).
solve(_____).
```